

Quantum computing is a rapidly-emerging technology that harnesses the laws of quantum mechanics to solve problems too complex for classical computers.

Today, IBM Quantum makes real quantum hardware -- a tool scientists only began to imagine three decades ago -- available to thousands of developers. Our engineers deliver ever-more-powerful superconducting quantum processors at regular intervals, alongside crucial advances in software and quantum-classical orchestration. This work is driving toward the quantum computing era of utility -- a period during which quantum computers can serve as a scientific tool to explore new classes of problems in chemistry, physics, and materials science, well beyond the reach of brute-force classical simulation.

What is a quantum computer? Classical computers, which include smartphones and laptops, encode information in binary "bits" that can either be 0s or 1s. In a quantum computer, the basic unit of memory is a quantum bit or qubit. Qubits are made using physical systems, such as the spin of an electron or the orientation of a photon.

These systems can be in many different arrangements all at once, a property known as quantum superposition. Qubits can also be inextricably linked together using a phenomenon called quantum entanglement. The result is that a series of qubits can represent different things simultaneously.

For instance, eight bits is enough for a classical computer to represent any number between 0 and 255. But eight qubits is enough for a quantum computer to represent every number between 0 and 255 at the same time. A few hundred entangled qubits would be enough to represent more numbers than there are atoms in the universe.

This is where quantum computers get their edge over classical ones. In situations where there are a large number of possible combinations, quantum computers can consider them simultaneously. Examples include trying to find the prime factors of a very large number or the best route between two places.

However, there may also be plenty of situations where classical computers will still outperform quantum ones. Therefore the computers of the future may be a combination of both these types.

The principles of quantum physics that quantum computing is based upon: Superposition: This is the ability of a quantum system to be in multiple states simultaneously. The go-to example of superposition is the flip of a coin, which consistently lands as heads or tails—a very binary concept. However, when that coin is in mid-air, it is both heads and tails and until it lands, heads and tails simultaneously. Before measurement, the electron exists in quantum superposition.

Entanglement: This is a strong correlation that exists between quantum particles — so strong, in fact, that two or more quantum particles can be inextricably linked in perfect unison, even if separated by great distances. The particles are so intrinsically connected, they can be said to "dance" in instantaneous, perfect unison, even when placed at opposite ends of the universe. This seemingly impossible connection inspired Einstein to describe entanglement as "spooky action at a distance."

Decoherence: This occurs when the quantum state of a qubit collapses due to interaction with its environment. This is one of the biggest challenges in building practical quantum computers, as qubits are incredibly sensitive to heat, electromagnetic fields, and other external disturbances.